Role of Heat Treatment in Designing Advanced Steels(QP,DP,TRIP)

MSE 307 : Materials Processing

Kanhaiyalal(23110155)
Raavi Hari Charan Teja(23110260)
Verkala Tanuj Kumar(23110358)

Introduction

Role of Heat Treatment in Advanced Steels

The Goal: Modern vehicles need to be Stronger (for safety) yet Lighter (for fuel efficiency).

The Challenge: The traditional trade-off: increasing strength reduces ductility.

The Solution: Advanced Heat Treatment

- Controls thermal paths to engineer specific microstructures.
- Combines soft phases (forming) with hard phases (strength).

Topics We Will Cover:

- **DP Steels:** Ferrite + Martensite (The "Workhorse").
- **TRIP Steels:** Using Retained Austenite for ductility.
- **Q&P Steels:** Carbon partitioning (3rd Generation).

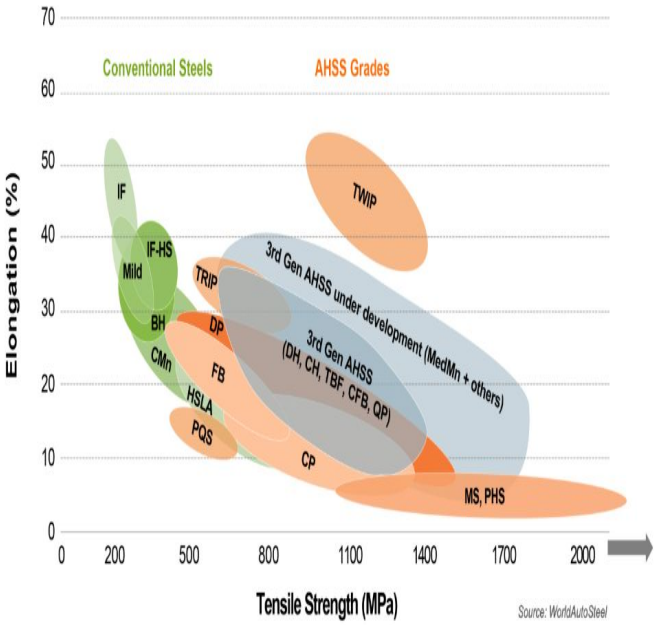
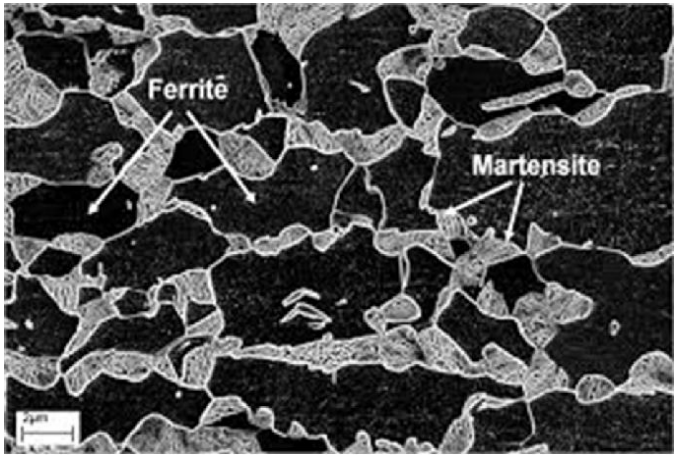


Figure 1: Global formability diagram showing the strength-ductility balance of AHSS.

Introduction to DP Steel

What is DP Steel?

DP steel is an advanced high-strength steel made of a **soft ferrite matrix** with **hard martensite islands**. This dual-phase design gives the steel **high strength, good ductility, and smooth yielding**, making it ideal for automotive crash-safety structures.



Key Features – Why It Works

- **Ferrite (soft phase):** provides ductility and formability
- **Martensite (hard phase):** gives high strength
- **Dual-phase structure:** combines low yield strength with high UTS

Common Applications

- Body panels
- B-pillar reinforcements
- Wheel housings

Figure 2: DP steel microstructure showing hard martensite (dark) dispersed in a soft ferrite matrix (light).

Key Properties & Microstructure

Microstructure

DP steel contains **martensite islands** dispersed in a **ferrite matrix**. This structure balances **strength + ductility**.

Key Features

- **Grain refinement** leads to better strength and ductility
- **Intercritical temperature** sets the ferrite/austenite ratio
- **Alloying additions (C, Mn, Si, Cr, Nb)** improve:
    - Hardenability
    - Martensite formation
    - Bake-hardening response
    - Grain refinement

Mechanical Balance

- Higher martensite = higher strength, lower ductility
- The **best balance** is typically at ~40–55% martensite.

Effect of Heat Treatment on DP Steels

How Heat Treatment Controls DP Steel Properties

Heat-treatment parameters such as temperature, time, and cooling rate directly set the ferrite/martensite ratio and mechanical behavior.

Key Heat-Treatment Effects

- **Annealing temperature:** controls ferrite/austenite distribution
- **Annealing time:** affects carbon diffusion

Cooling Rate Influence

- **Fast Quench:**Austenite becomes **Martensite** (Higher Strength)
- **Slow cooling:** Austenite becomes **Ferrite + Pearlite** (Lower Strength)
- The quench must be fast enough “beat” pearlite formation.

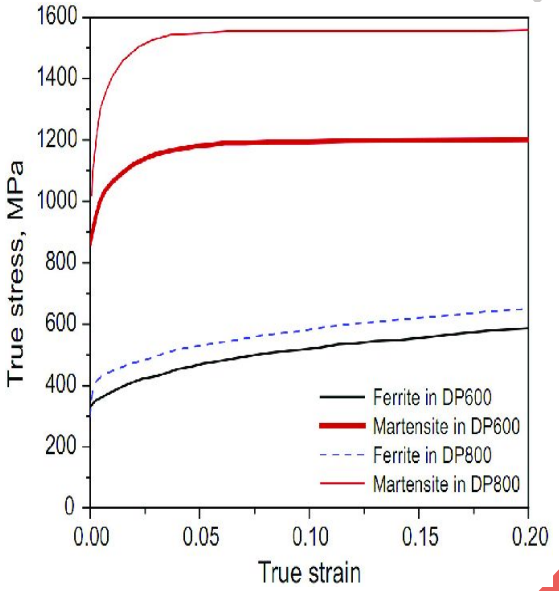


Figure 3: Stress-strain curves of DP steels exhibiting continuous yielding and high work hardening.

Recent Advances & Applications

Recent Advances in DP Steel Development

Objective: Optimize thermal cycles & alloying for superior balance.

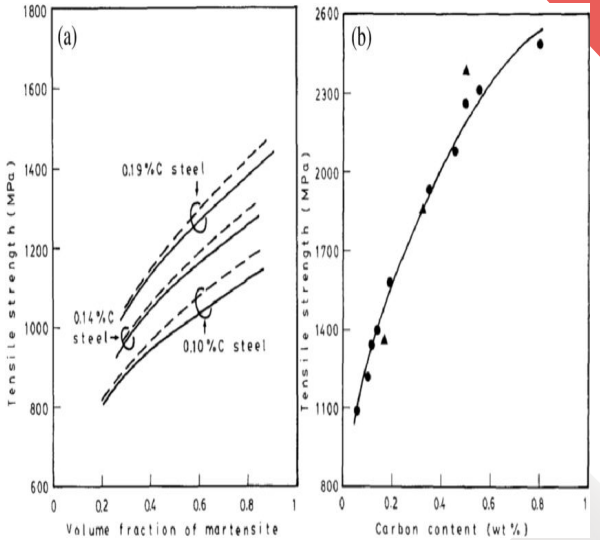
Key Advances

- **Precise Thermal-Cycle Control:** "Tuning" the microstructure.
- **Nb-Microalloying:** Grain refinement for higher strength.

Applications & Industry Use

- Used in AHSS grades **500–1200 MPa**
- **The "Safety Cage":** side members, B-pillars, wheel housings, floor reinforcements

Figs. 4 & 5: Diagrams showing the effect of martensite volume fraction (left) and carbon content (right) on tensile strength.



TRIP Steel & Microstructure

What is TRIP Steel?

TRIP steel is an advanced steel where retained austenite (RA) transforms into martensite when the material is deformed. This **TRIP effect** gives the steel a rare combination of high strength and high ductility, which is why it's widely used in automotive crash-safety parts and energy-absorbing components.

Microstructure (Key Phases) – Why It Works

- **Ferrite:** soft phase that provides ductility and easy formability
- **Bainitic ferrite:** stronger phase that adds strength and supplies carbon
- **Retained austenite:** transforms under strain and increases strength during deformation
- **Martensite (less):** adds initial strength to the steel

Fig1: shows TEM micrograph of the C-Mn-Al-P trip steel(Ding et al., JMR&T).

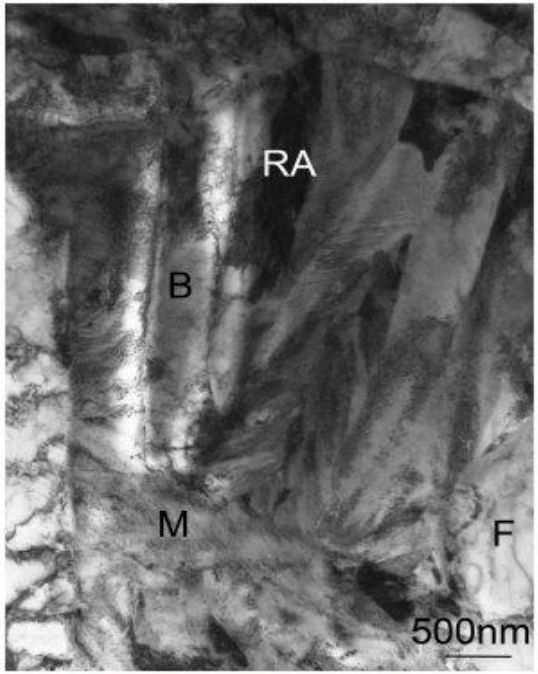


Fig. 3. TEM micrograph of the C–Mn–Al–P steel.

Mechanisms & Kinetics

Mechanisms – How phases transform

- **Heat Treatment:** Austenite becomes bainitic ferrite, and carbon moves into retained austenite, making it stable.
- **Deformation:** Retained austenite instantly turns into martensite, giving extra strength its the TRIP effect.

Kinetics – How fast it happens

- Bainite forms very fast in the first 0–100 seconds, then slows down.
- 480°C IBT: faster early transformation.
- 440°C IBT: slower but more controlled.

Why It Matters

- The transformation speed decides how stable the retained austenite becomes.
- More stable RA, better ductility and a stronger TRIP effect.

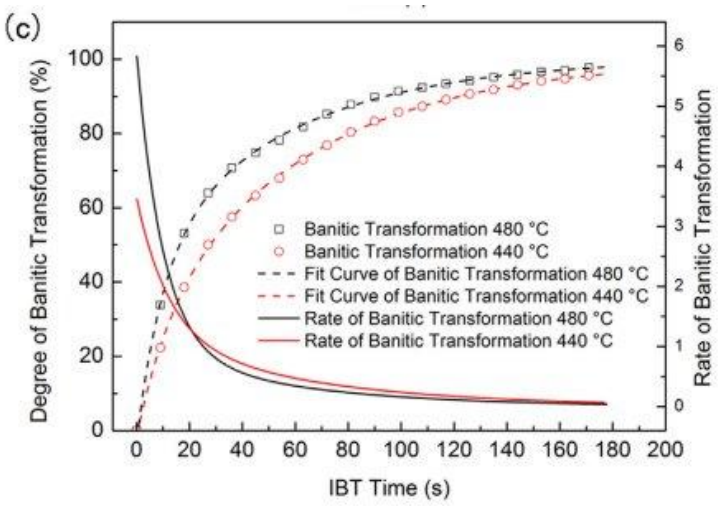


Fig 2: Ding et al., JMR&T — Bainite transformation at 440°C & 480°C.

Thermodynamics & Heat-Process Factors

Thermodynamics – Why transformations occur

- Carbon goes into austenite and makes it stable
- Ms temperature becomes lower than room temperature
- Retained austenite remains after cooling and transforms during strain

Key Heat-Treatment Factors

- IA at 820–880°C forms ferrite and austenite.
- IBT time is most important
  - Long IBT time gives stable retained austenite
  - Short IBT time gives more martensite
- IBT temperature (440–480°C) controls how fast bainite forms
- C and Mn help stabilize austenite; Al and P prevent carbides

Fig 3 shows how retained austenite increases and martensite decreases with IBT time (Ding et al., JMR&T).

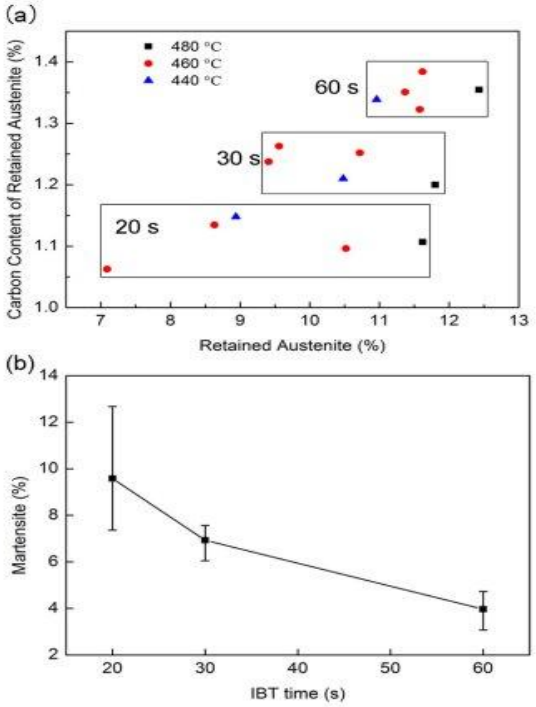


Fig. 7. Influence of IBT time and temperature on retained austenite fraction and carbon content (a) and influence of IBT time on martensite fraction (b).

Effect of Mechanical & Thermal Conditions

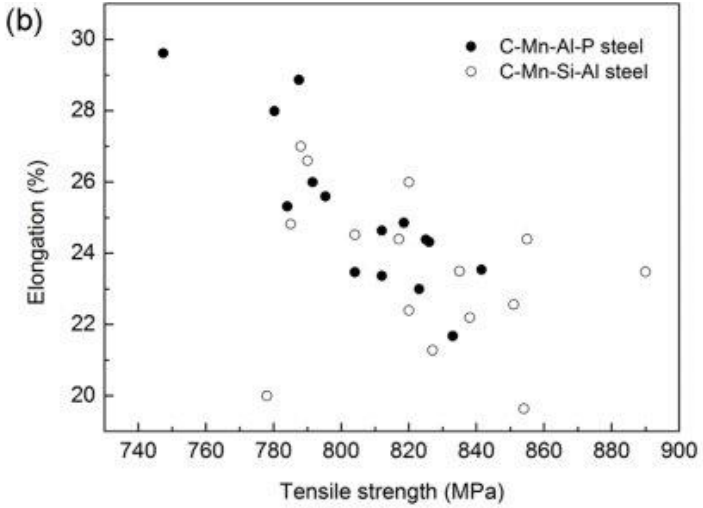
Mechanical Effects (Stretching)

- Retained austenite slowly turns into martensite during deformation
- More retained austenite gives **better ductility**
- More martensite increases strength but **reduces ductility**

Thermal Effects (During forming)

- Low temperature: austenite transforms easily → higher strength, lower ductility
- High temperature: austenite stays stable → higher ductility

Fig 4 shows strength and ductility of the C–Mn–Al–P TRIP steel and the reference steel (Ding et al., JMR&T).



Quenching & Partitioning (Q&P) Steels

- We have looked into DP and TRIP steels. Now let's look into QP steels.
- **Q&P Goal:** Get a super-strong Martensite matrix + the TRIP effect from Retained Austenite (γ_R).
- **The Q&P Idea:**
  - **Quench:** Partial quench from A_T to Q_T (between M_s and M_f) → Martensite(α') + Austenite(γ).
  - **Partition:** Hold at P_T → Carbon moves (partitions) from Martensite to Austenite.
  - **Final Quench:** This is where the Carbon-rich Austenite becomes stable Retained Austenite.

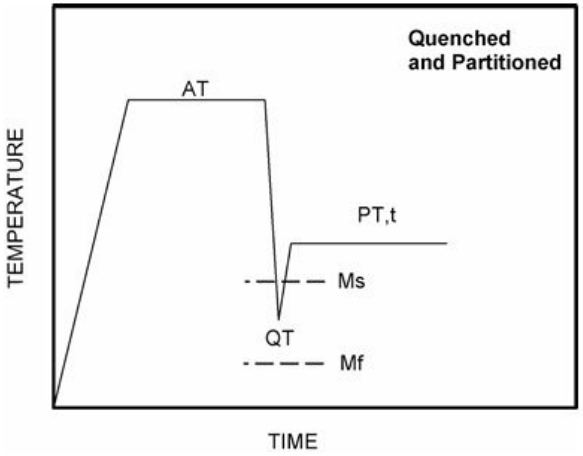


Fig. 1. QP Temp-Time graph

Mechanisms Inside QP Steels

- **What actually happens inside:**
  - Initial quenching forms supersaturated martensite + untransformed austenite.
  - During the partitioning, carbon diffuses from martensite into austenite.
  - Because of this carbon enrichment, austenite becomes stable retained austenite at room temperature.
  - On final cooling, some fresh martensite may form if some austenite wasn’t enriched enough.
  - To delay the formation of carbides, we use silicon as carbide-suppressing element.

Thermodynamics & Kinematics of QP Steels

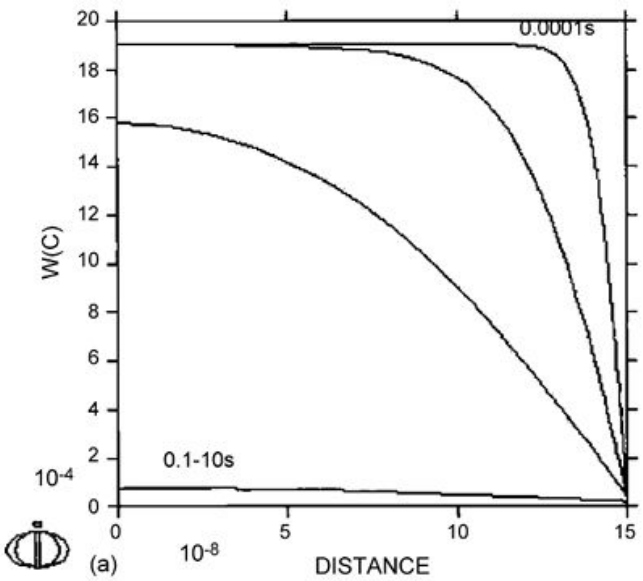
Thermodynamics:

- The flow of carbons is due to the chemical difference between martensite and austenite.
- Here, partitioning aims for CCE (equal μC in both phases).
- In this, to suppress the formation of carbides, we use silicon to raise cementite free energy.
- Higher P_T = stronger driving force, but risk of bainite/carbides.

Kinematics:

- Carbon diffuses very fast at 350–450°C (in seconds).
- Here, transformation interface stays mostly stationary, which means only C moves.
- The carbon transfer from martensite to austenite happens in an instant, but austenite homogenization takes a little longer.

Thermodynamics & Kinematics of QP Steels



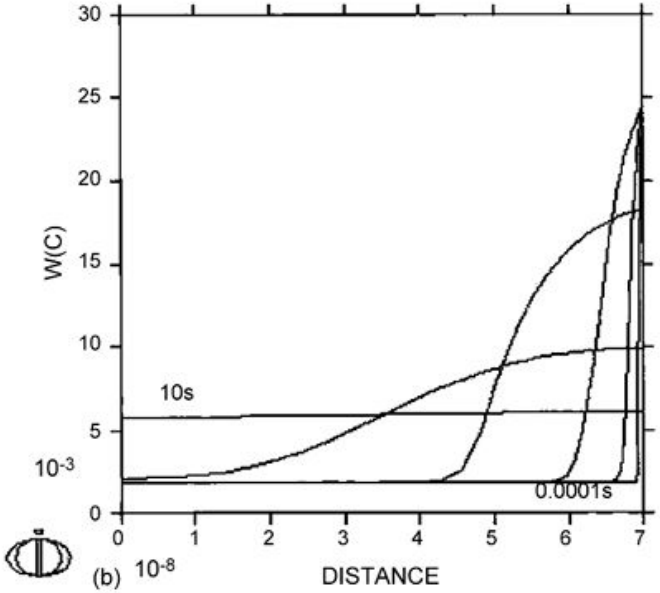


Fig.2a & 2b. Snapshots of transfer of carbon from martensite (2a) to austenite (2b) at specific time intervals

Microstructure Of QP Steels

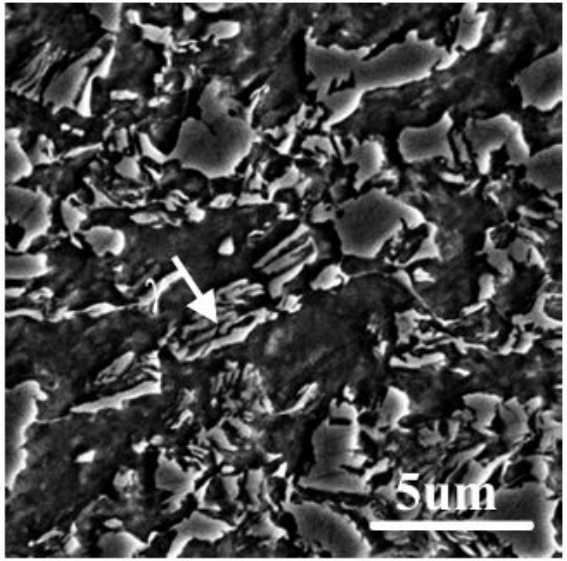


Fig. 2. Final Q&P Microstructure (SEM)

Resulting Microstructure:

- Final structure is a mixture of lath martensite + carbon-enriched retained austenite.
- Retained austenite becomes stable because carbon partitions into it during the partitioning step, lowering its M_s.
- Martensite gives high strength, while retained austenite provides ductility through the TRIP effect during deformation.
- If partitioning temperature/time is high, a small amount of bainite may form. This also contributes to carbon enrichment but can reduce retained austenite fraction.
- The balance between martensite, retained austenite, and fresh martensite strongly affects strength, ductility, and strain hardening behaviour.

Key Controllers Of QP Heat Treatment

We can tune the final microstructure by 3 main factors.

1. **Quench Temperature (Q_T):**
   - This controls the initial fraction of α'/γ (martensite/austenite).
   - We must find the "sweet spot"/Optimum Q_T to maximize retained austenite (γ_R).
2. **Partitioning Temp/Time (P_T, t):**
   - This controls the kinematics of C diffusion vs. other reactions.
3. **Alloying (C, Si):**
   - Carbon is the fuel that powers the QP mechanism.
   - Silicon is used to delay the formation of carbides.

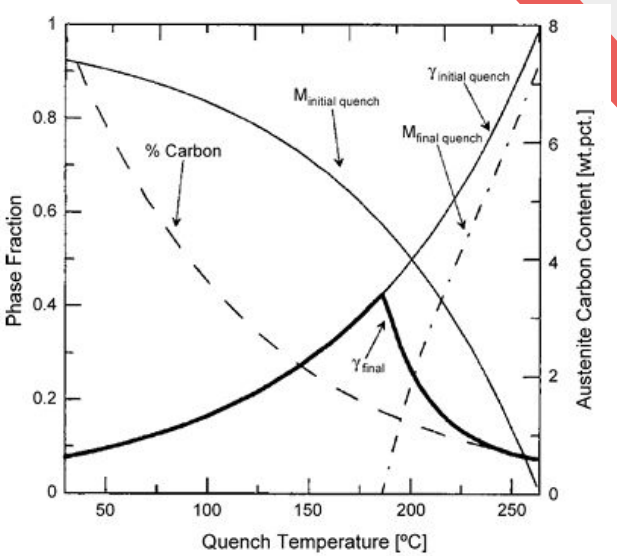


Fig. 4: Predicted Final Phase Fractions vs. Quench Temperature (Q_T)

Properties & Applications Of QP Steels

- **Structure-Property Relationship:**
  - High Strength (from martensite, >1GPa).
  - High Ductility (from retained austenite, TRIP effect).
- The combination of high strength and ductility makes QP steels very effective at absorbing impact energy.
- During deformation, retained austenite transforms and work-hardens the material, allowing it to deform in a controlled way rather than failing suddenly.
- Due to this, QP steels are widely used in next-generation automotive structures such as B-pillars, rocker panels, reinforcements, and other components where both high strength and energy absorption are essential.

Comparison Table

Feature	QP Steel	DP Steel	TRIP Steel
Microstructure	Martensite + retained austenite	Ferrite + martensite	Ferrite + bainite + RA + Martensite (very less)
Strength	Highest (martensite-rich)	Medium–high	High
Ductility	Moderate-high (via stable RA)	Moderate	Highest (strong TRIP)
Key Mechanism	Carbon partitioning, which leads to retained austenite stability	Load sharing between ferrite and martensite	Strain-induced martensite (TRIP effect)
Processing	Fast (no bainite hold)	Simple & fast	Slow due to bainite step

Future Scope & Industrial Relevance

Future Scope

- Improve heat-treatment design for better strength–ductility balance
- Develop next-generation DP, TRIP, and QP steels with more stable retained austenite
- Use simulation and digital tools to speed up microstructure optimization

Industrial Relevance

- Used in lightweight and crash-safe automotive components
- Helps reduce vehicle weight, improving mileage and lowering emissions
- Works well with current industrial annealing and galvanizing lines

Fig 5 shows how automobiles use DP, TRIP, and QP steels for lightweight structure and high crash-safety (Cosasteel, 2023).

Conclusion

- Heat treatment lets us control phase transformations and design the microstructure we want.
- DP, TRIP, and Q&P steels each use different heat-treatment paths to balance strength and ductility.
- DP steels combine ferrite + martensite for a simple, strong, lightweight structure.
- TRIP steels use stable retained austenite for extra ductility and energy absorption.
- Q&P steels use quenching and carbon partitioning to achieve high strength with useful ductility.

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THANKS!